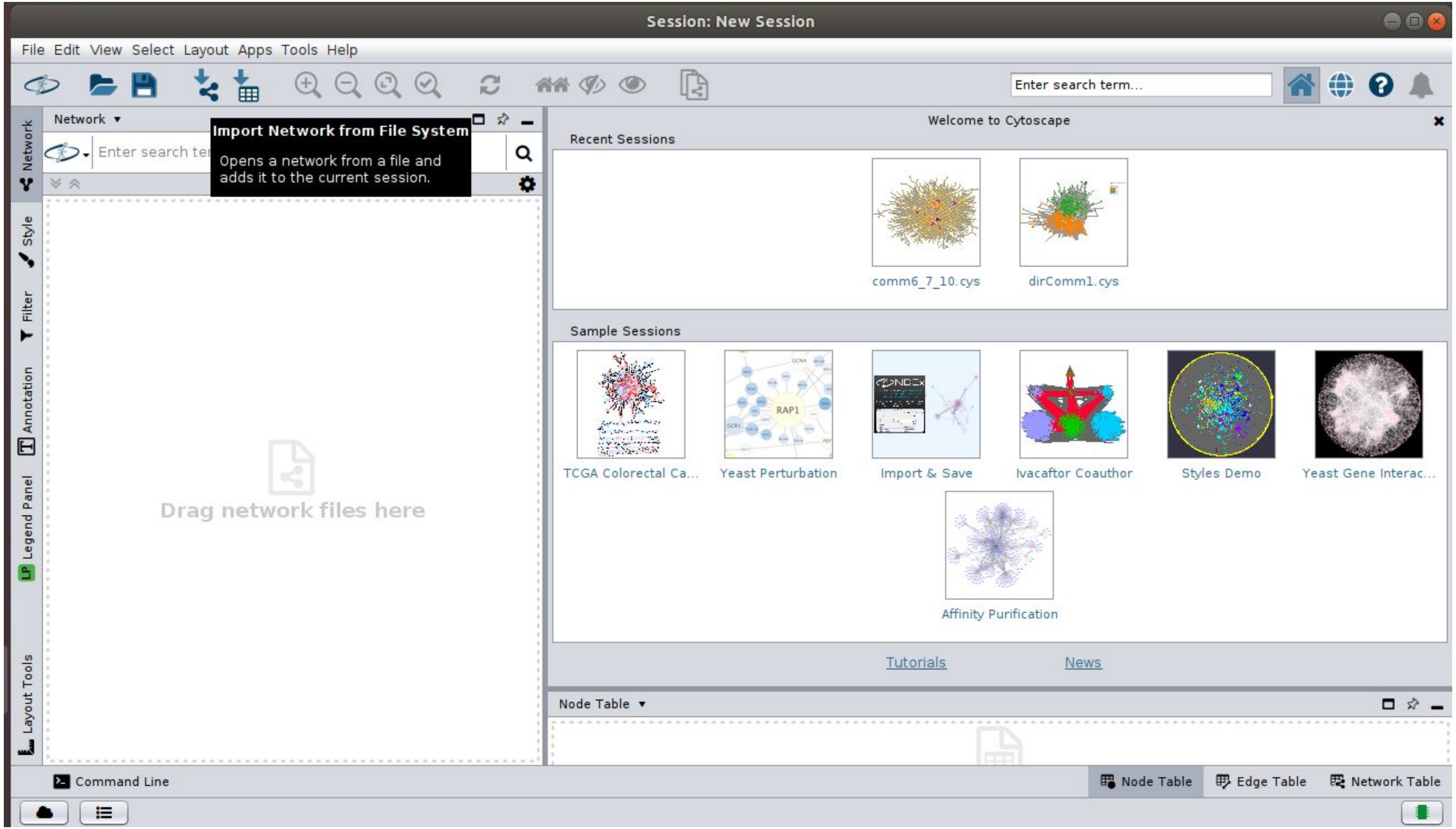
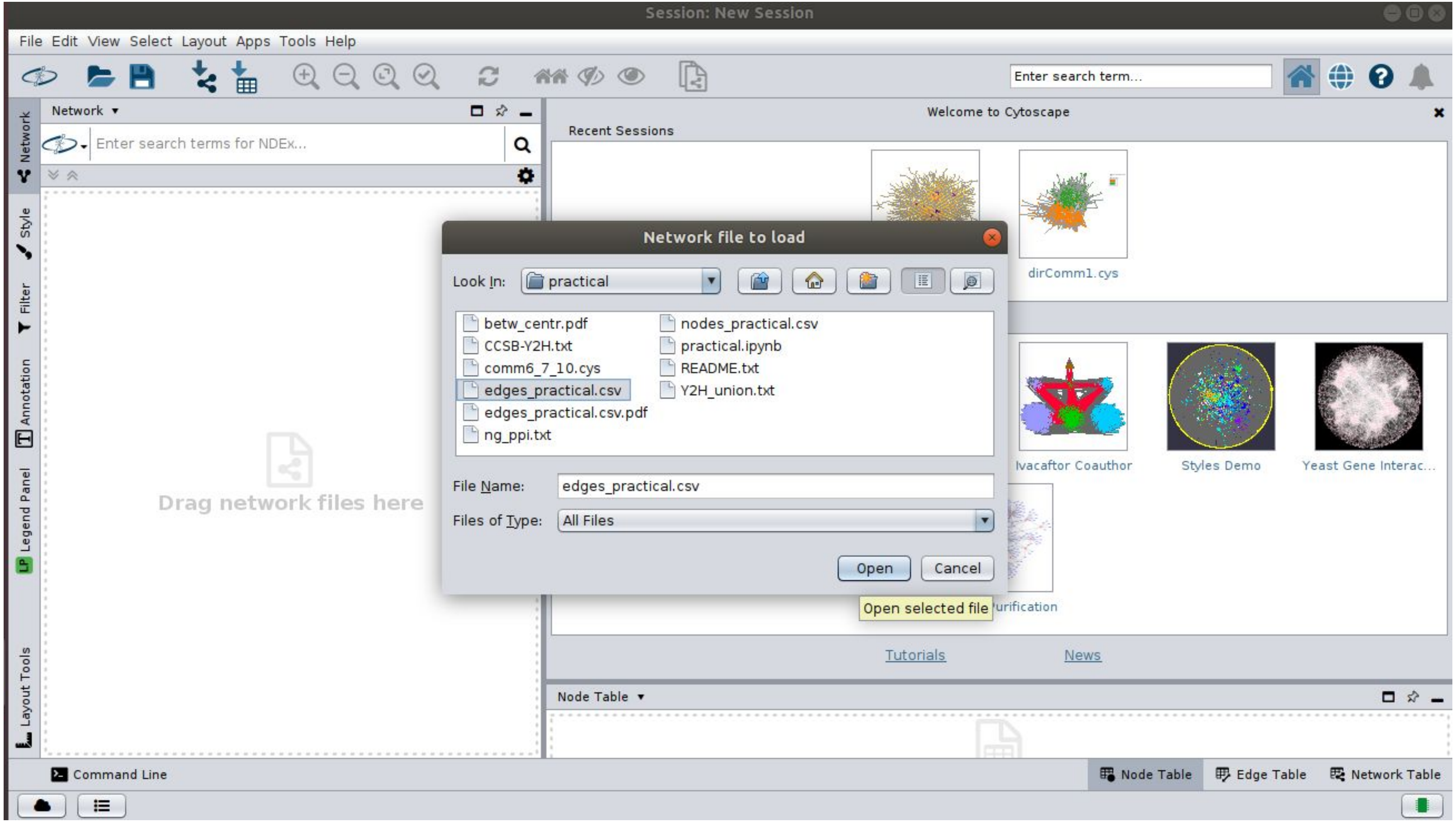
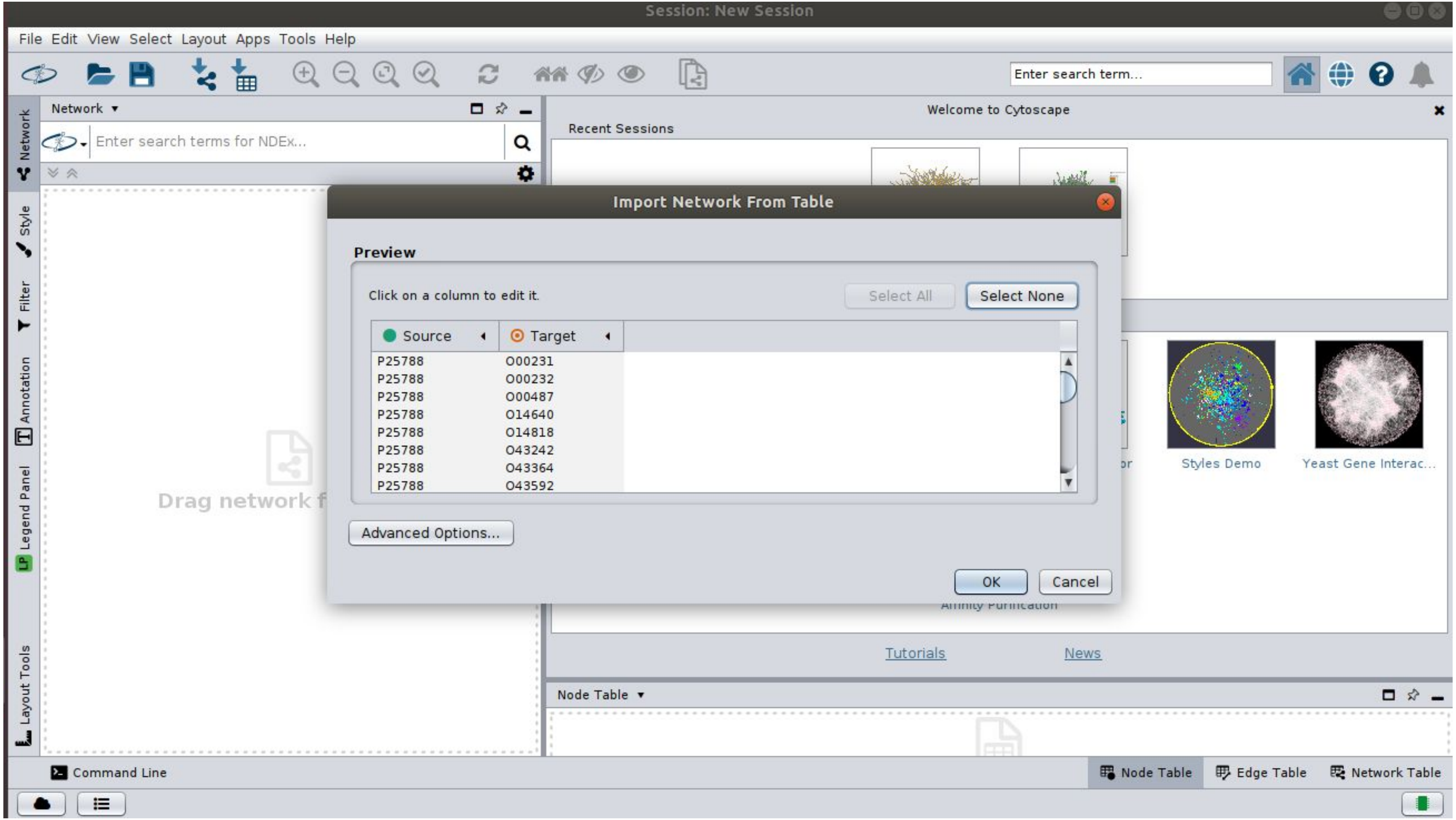
Cytoscape demo

Go to “**Import Network**” to import edge data for visualization

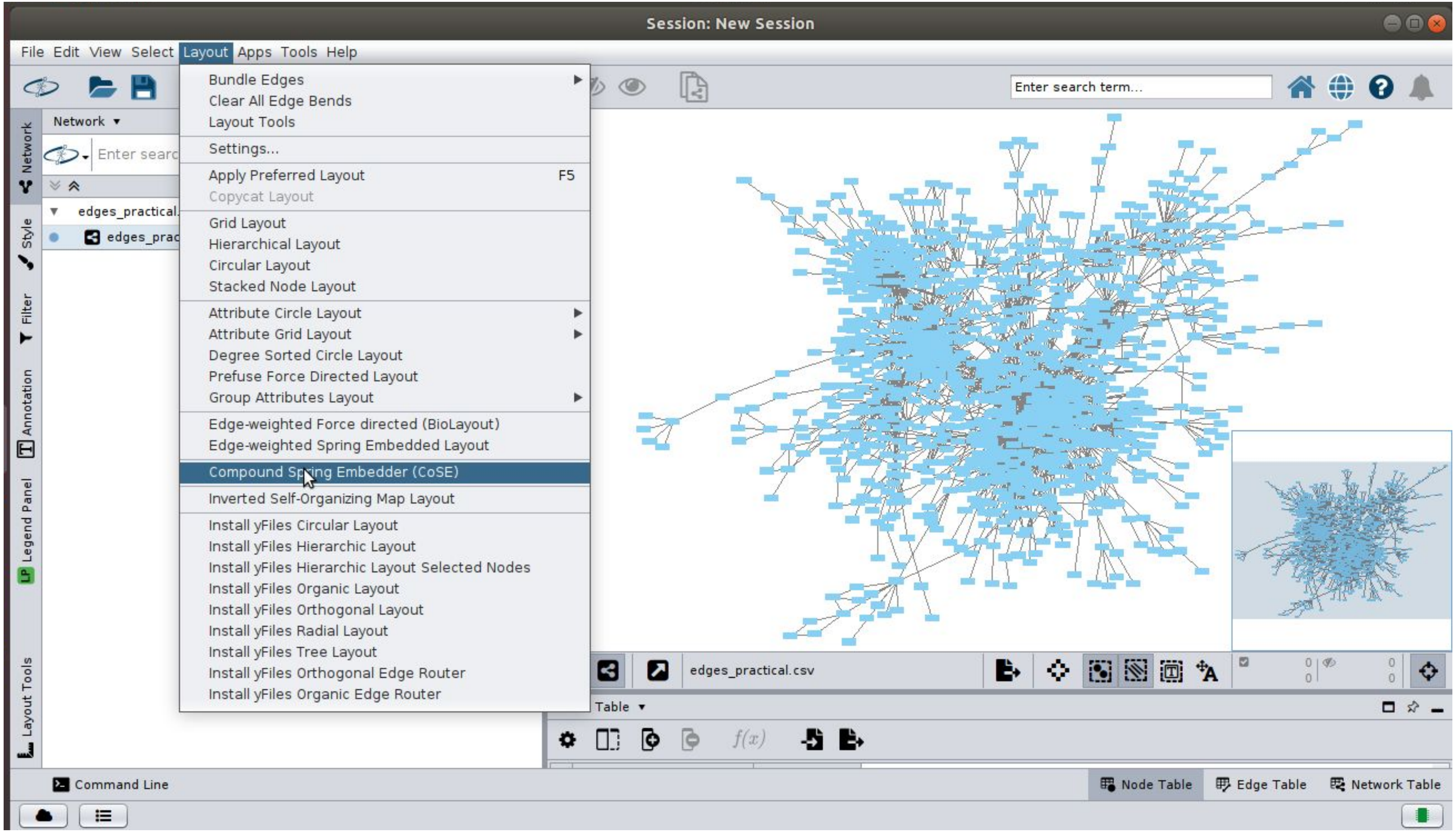


Select file containing your edgelist from the directory.

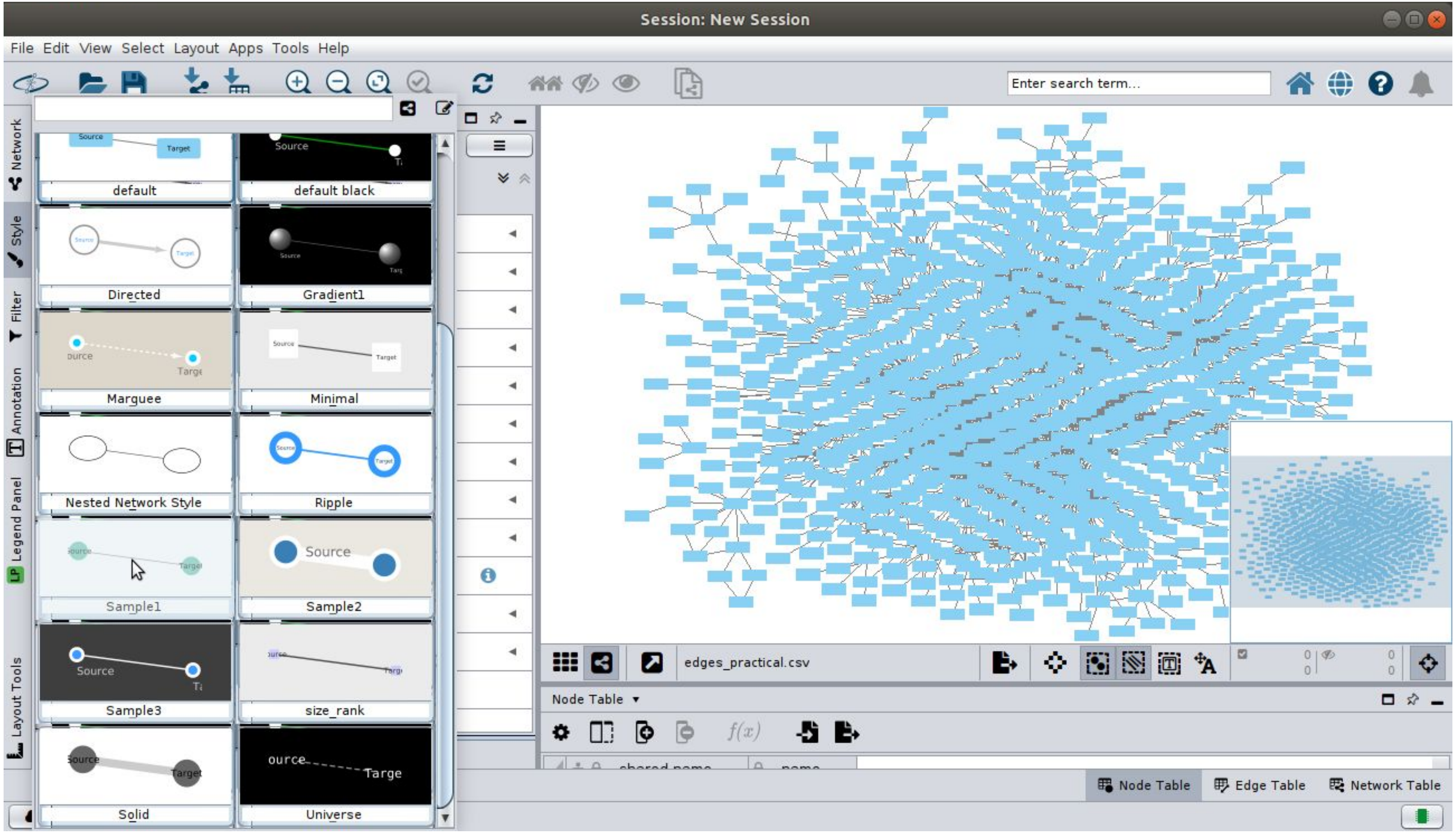



Session: New Session
File Edit View Select Layout Apps Tools Help
Enter search term...
Network
Enter search terms for NDEx...
Welcome to Cytoscape
Recent Sessions
Import Network From Table
Preview
Click on a column to edit it.
Select All
Select None
Source
Target
P25788 O00231
P25788 O00232
P25788 O00487
P25788 O14640
P25788 O14818
P25788 O43242
P25788 O43364
P25788 O43592
Advanced Options...
OK
Cancel
Drag network f
Styles Demo
Yeast Gene Interac...
Affinity Purification
Tutorials
News
Node Table
Command Line
Node Table
Edge Table
Network Table
Network
Style
Filter
Annotation
Legend Panel
Layout Tools

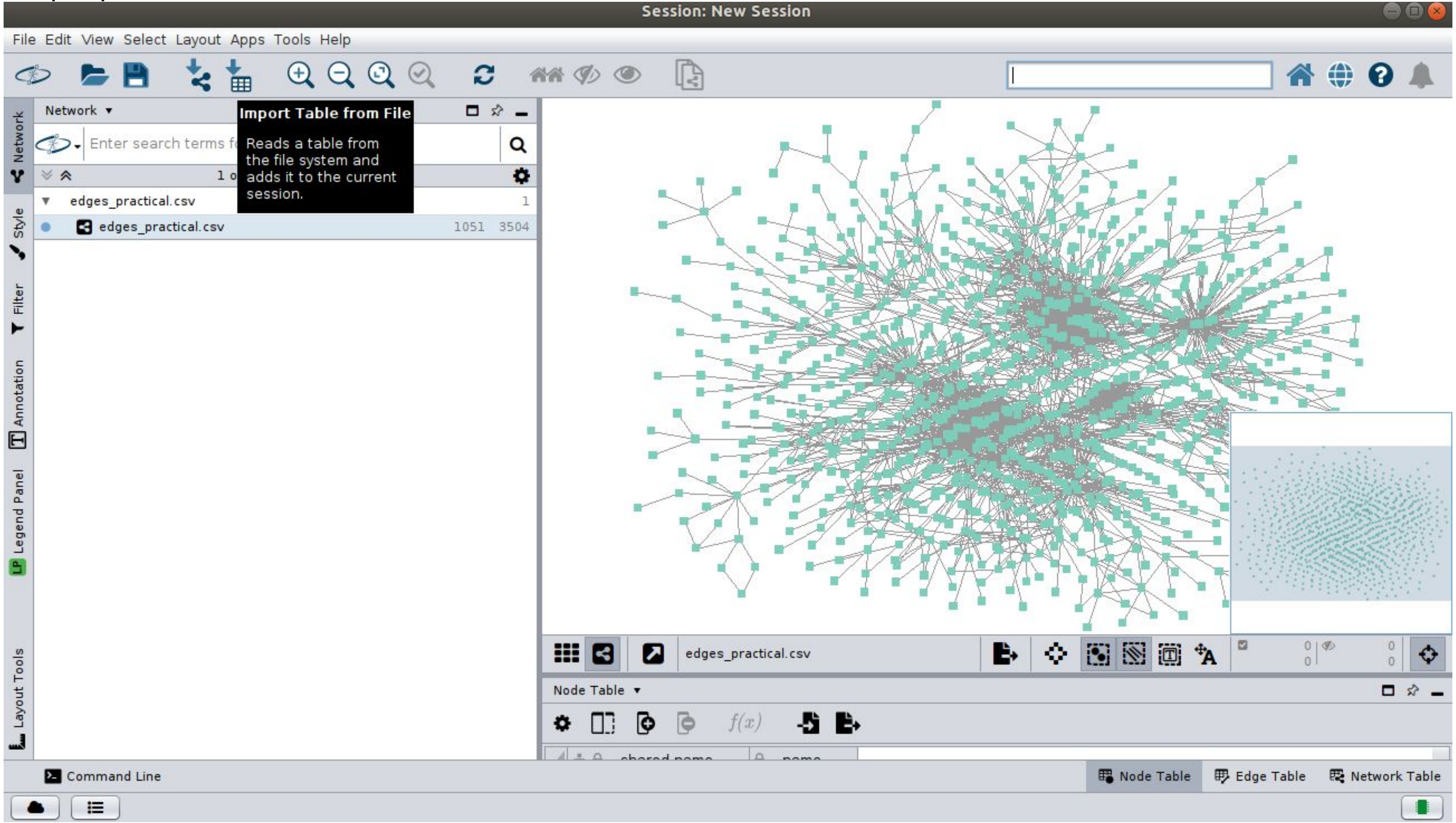
Change the “**Layout**” according to the one best suited for your visualization.



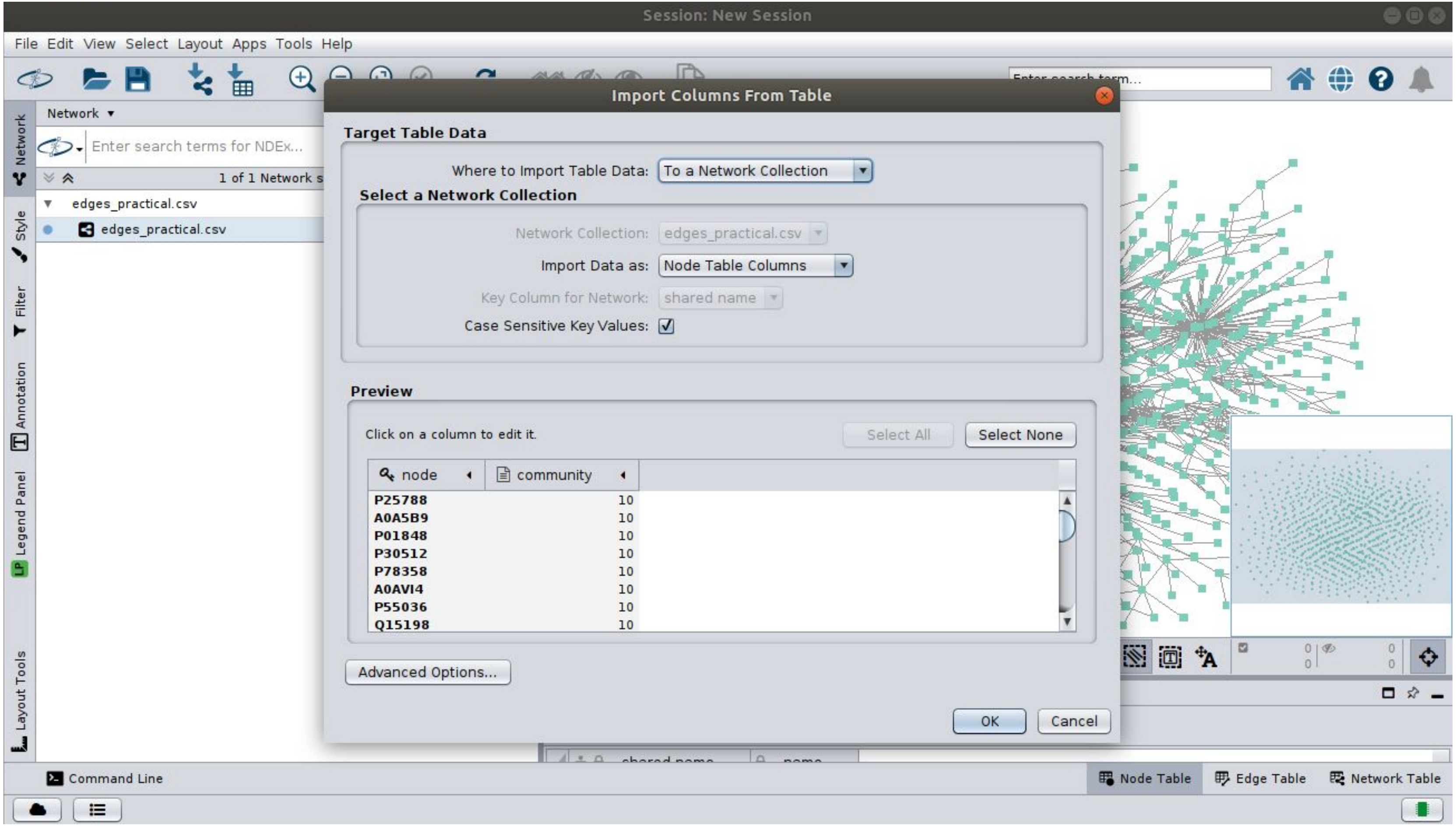
Select your network representation “**Style**” according to your preference



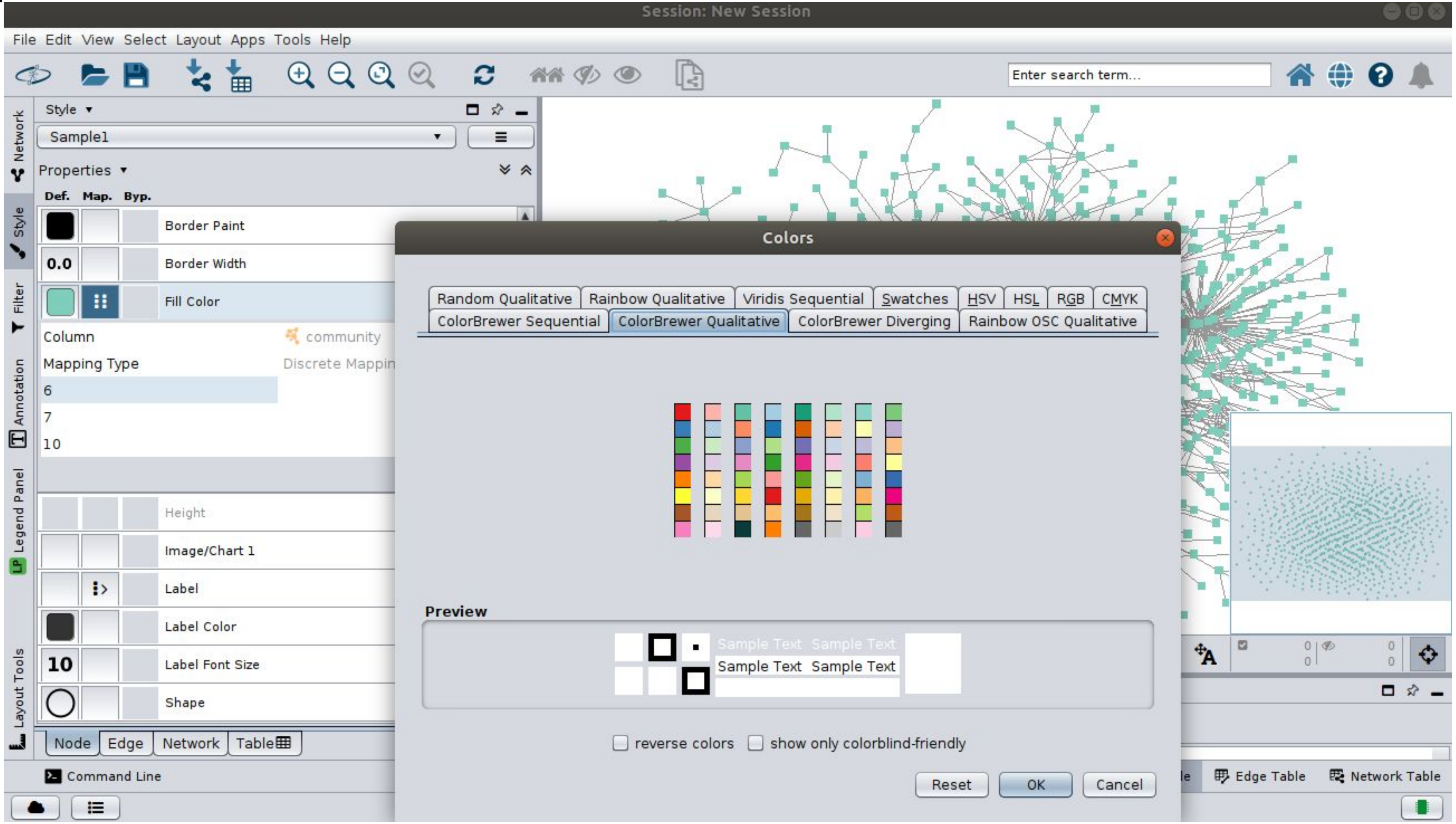
Import nodes file containing node attributes for visualizing certain node properties.



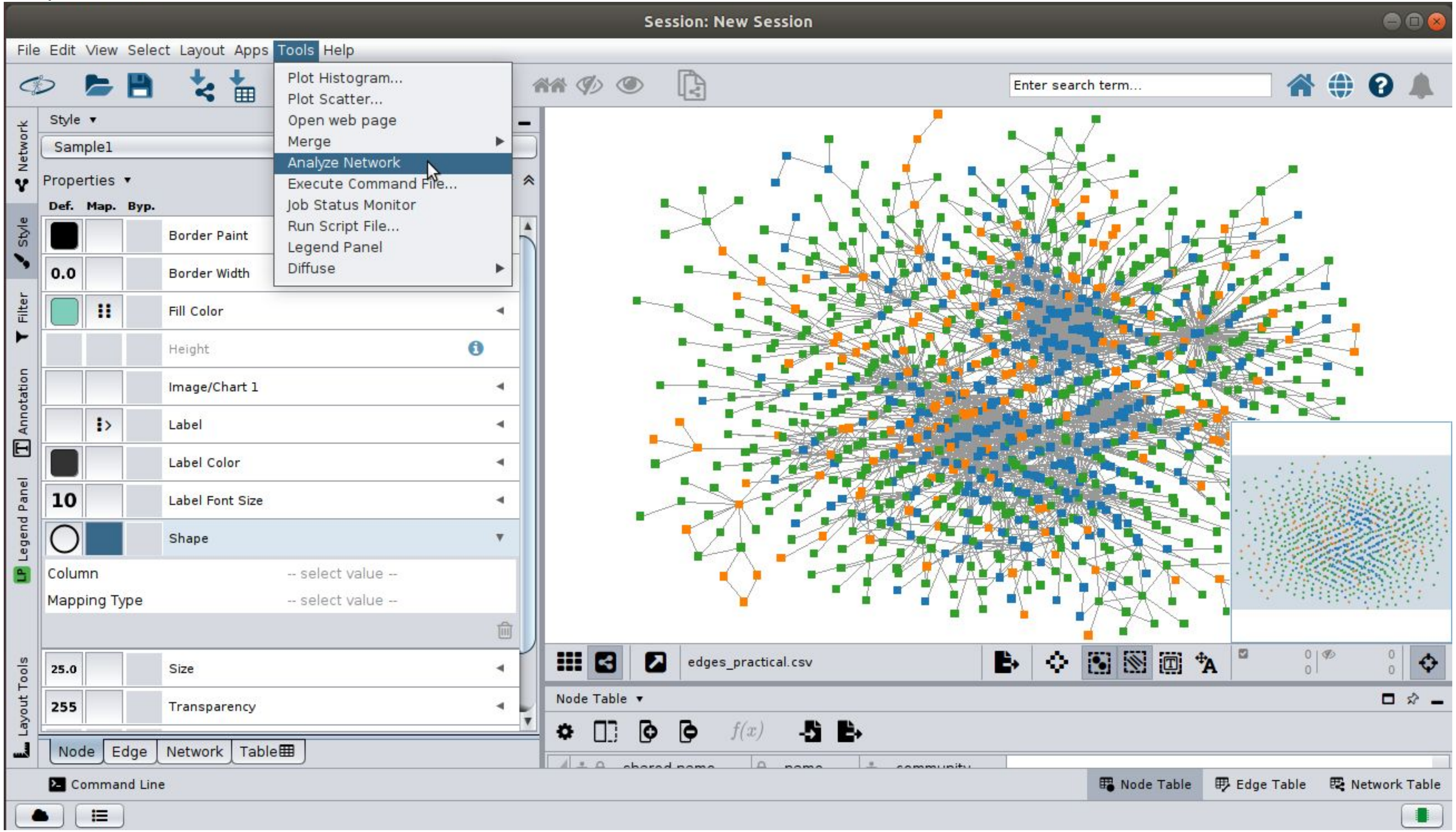
Make sure that “**Import data as:**” should be **Node Table Columns.**



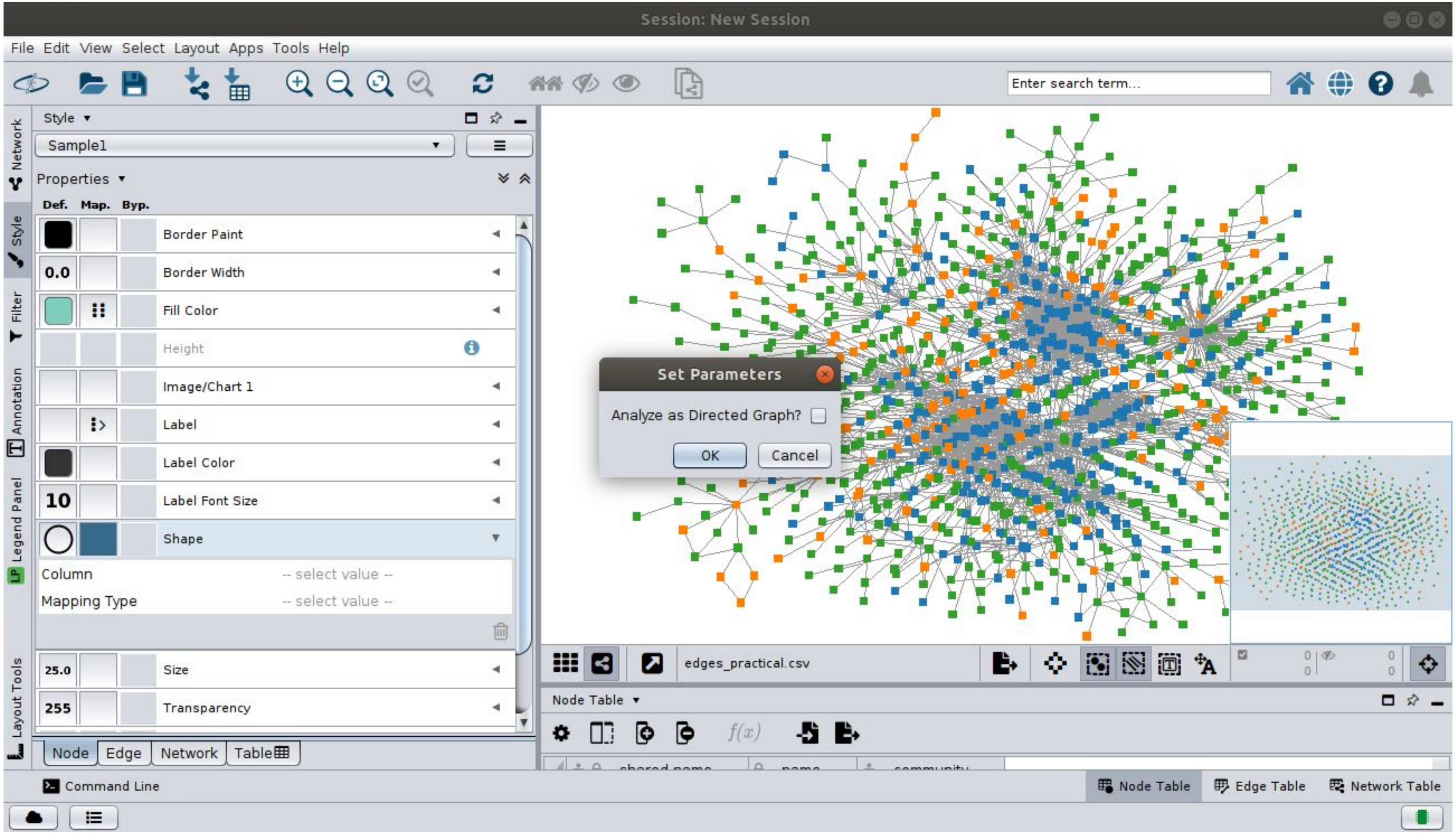
To color the nodes according to community, go to **Fill color** in **Styles**. Select **Column -> community**, **Mapping type -> Discrete** since the communities are only 3. Give unique color to each community according to your preference



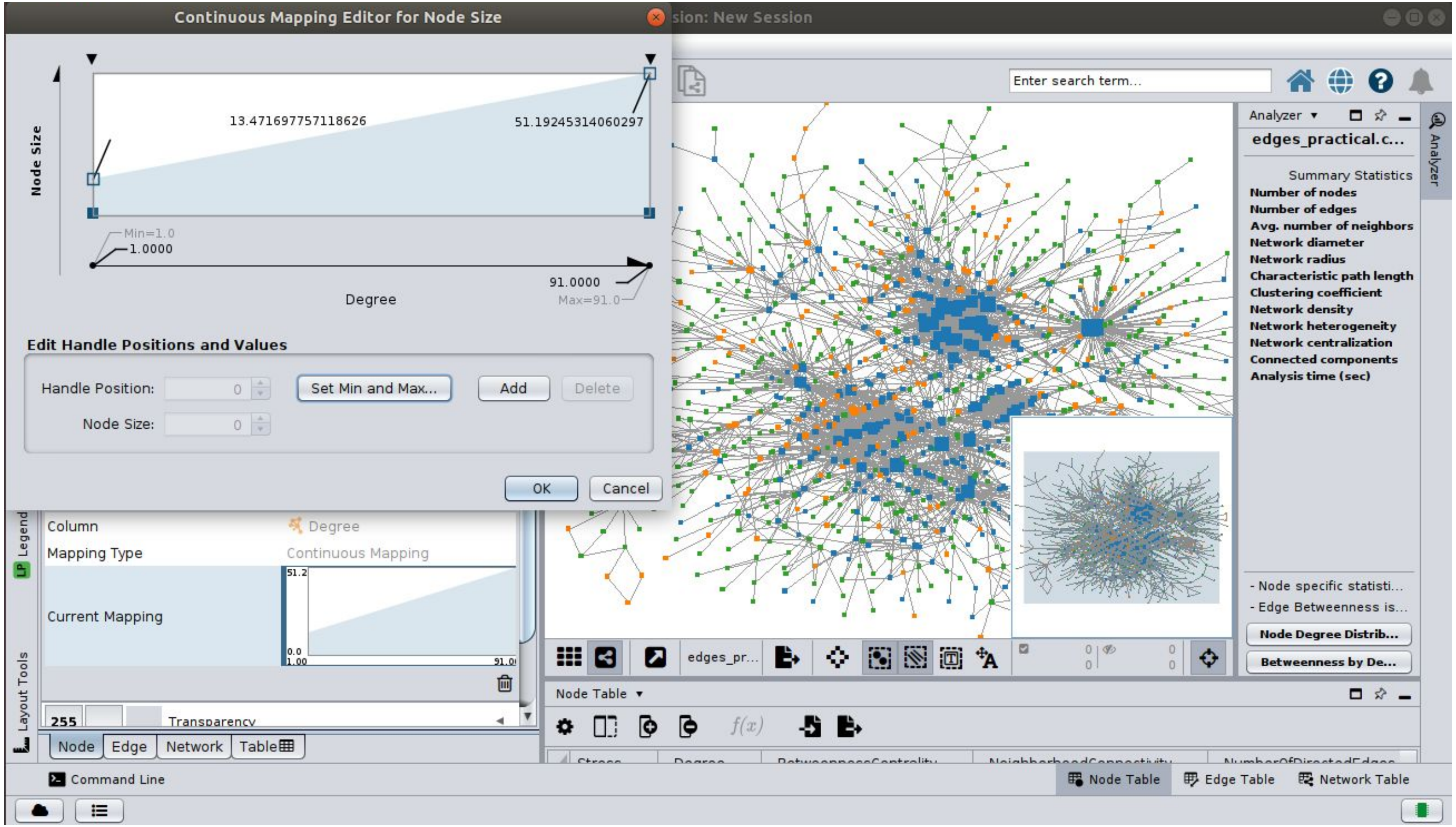
To represent the nodes according to various centralities like their degree, betweeness...., go to **Tools** and click on **Analyze Network**. All the properties of the network for each node are computed by the cytoscape software.



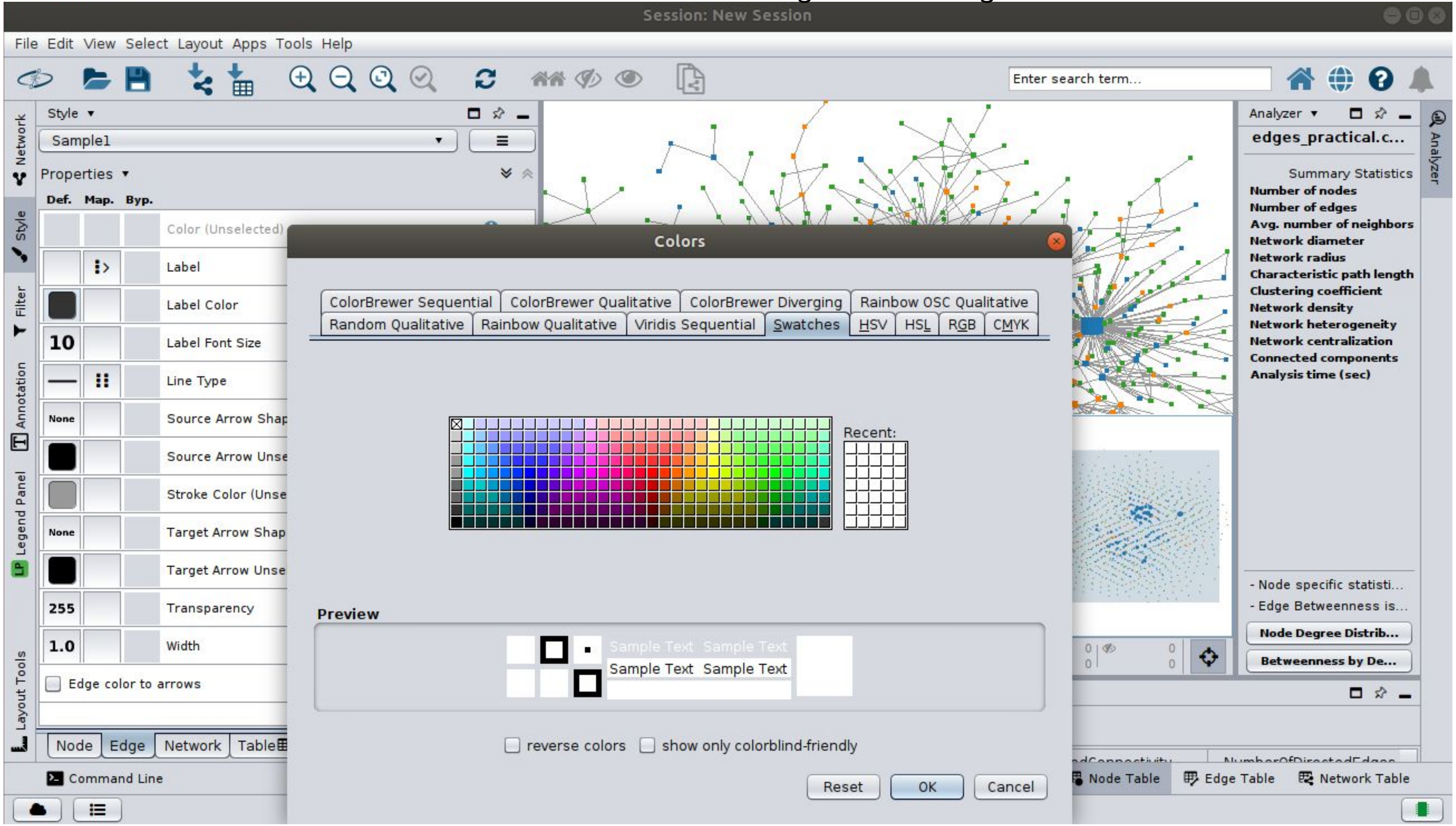
Do not select “Directed Graph” since our network is not directed.

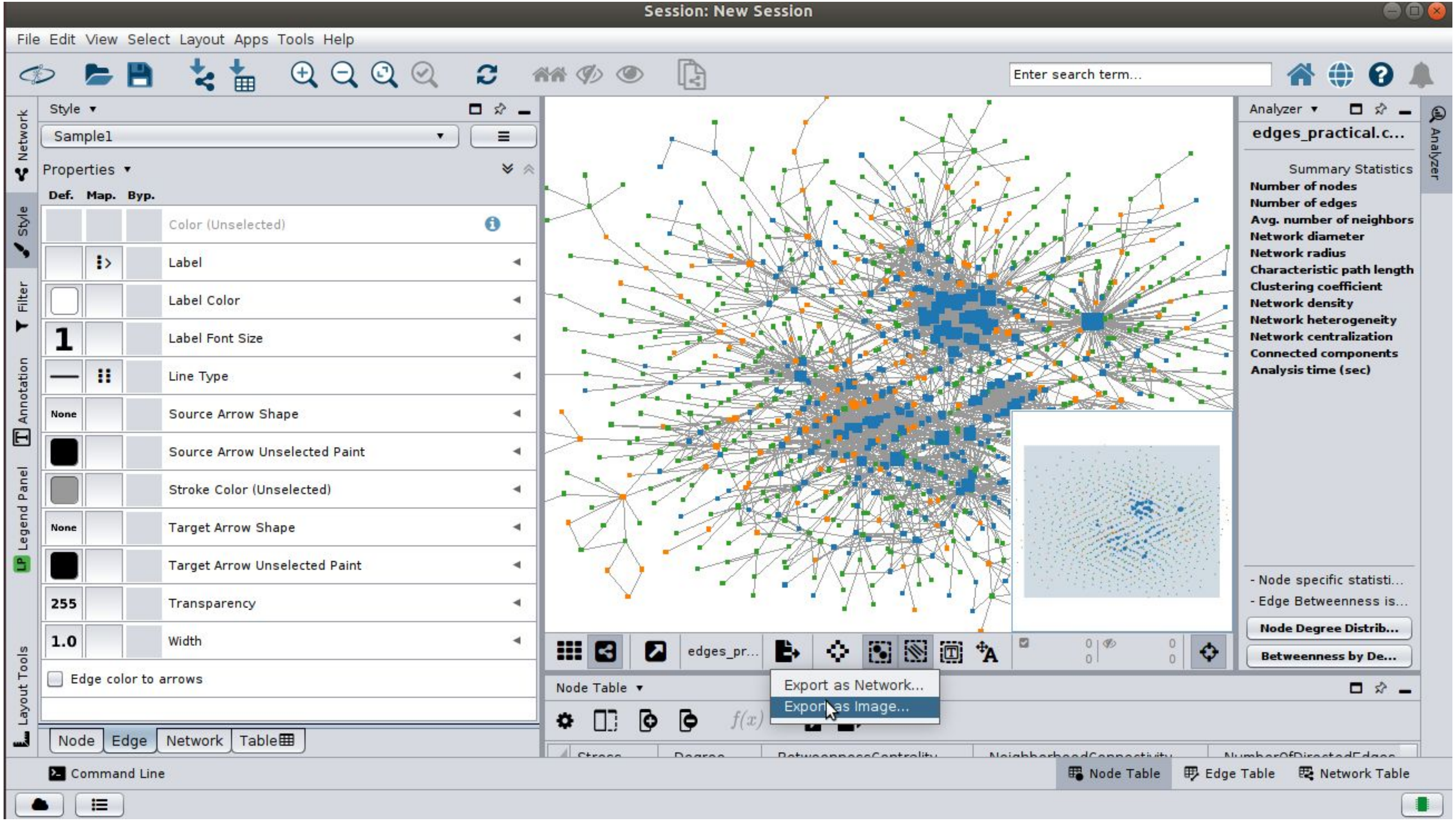


To represent the nodes according to degree, go to **Size** in **Styles**. Select **Column -> Degree**(which appears only after you **“Analyze network”** as mentioned in previous slides), **Mapping type -> Continuous.** Modify the range of the degree in the window to clearly observe that the size of high degree nodes should be bigger relatively to those of low degree nodes.



You can possibly try to remove the edge label by setting it to white in the **Label** option of **Edge** and also set **Label Font size** to 1 so that it doesn't interfere in the end image after saving.





Session: New Session
File Edit View Select Layout Apps Tools Help
Enter search term...
Style
Sample1
Properties
Def. Map. Byp.
Color (Unselected)
Label
Label Color
Label Font Size
Line Type
Source Arrow Shape
Source Arrow Unselected Paint
Stroke Color (Unselected)
Target Arrow Shape
Target Arrow Unselected Paint
Transparency
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None
1
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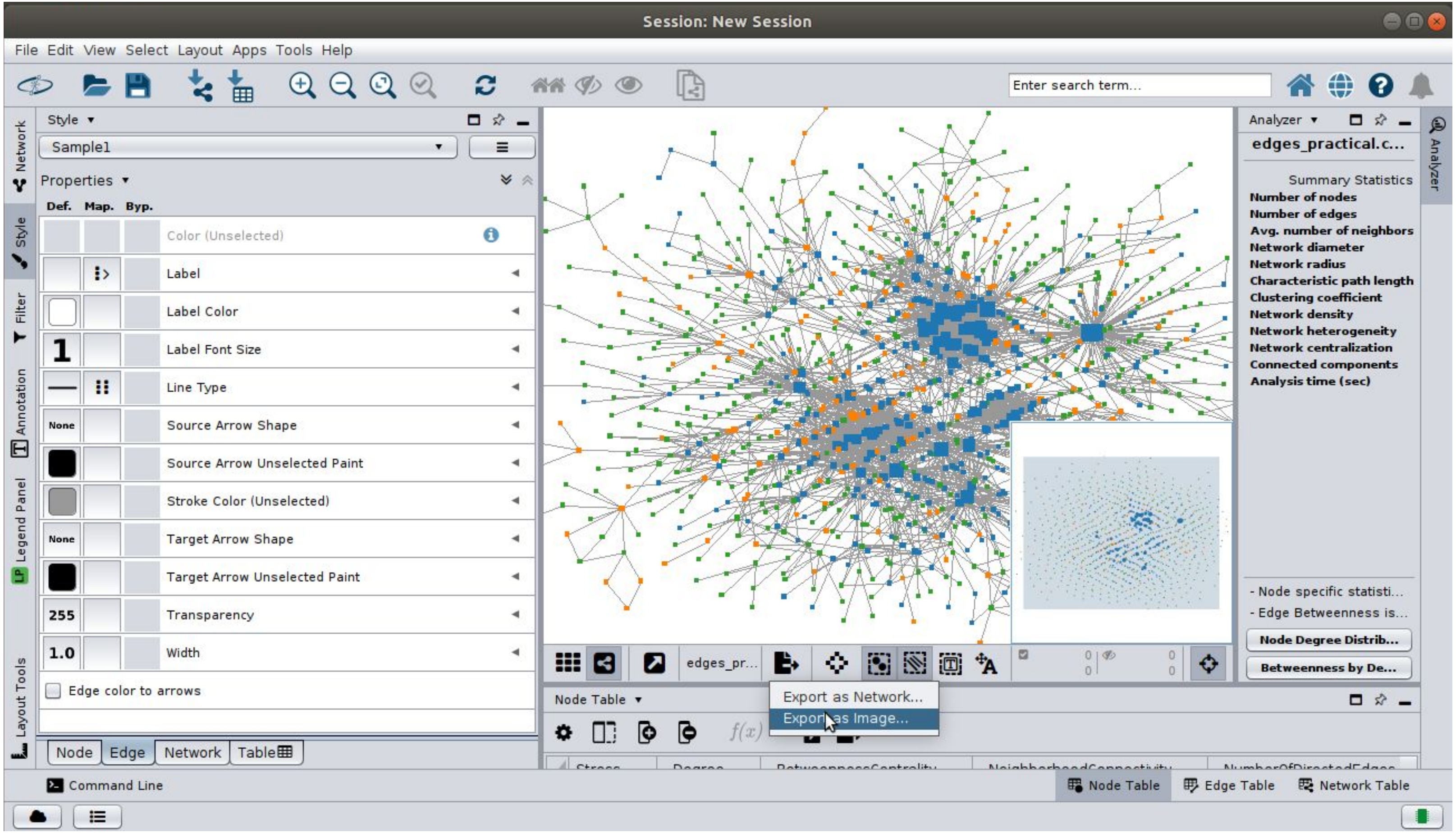
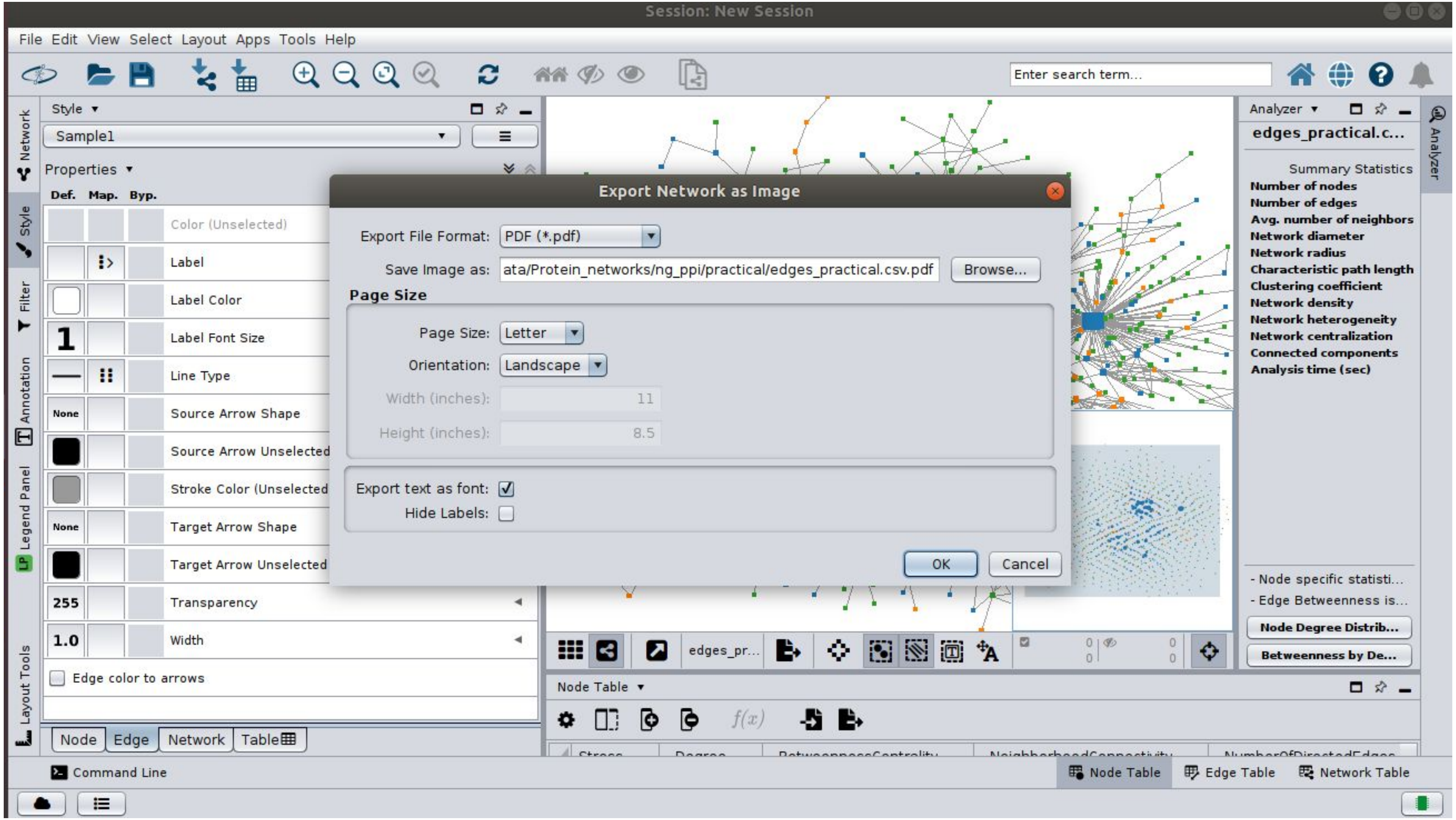
Edge color to arrows
Node Edge Network Table
Network
Style
Filter
Annotation
Legend Panel
Layout Tools
Analyzer
edges_practical.c...
Summary Statistics
Number of nodes
Number of edges
Avg. number of neighbors
Network diameter
Network radius
Characteristic path length
Clustering coefficient
Network density
Network heterogeneity
Network centralization
Connected components
Analysis time (sec)
- Node specific statisti...
- Edge Betweenness is...
Node Degree Distrib...
Betweenness by De...
edges_pr...
Export as Network...
Export as Image...
Node Table
Command Line
Node Table
Edge Table
Network Table

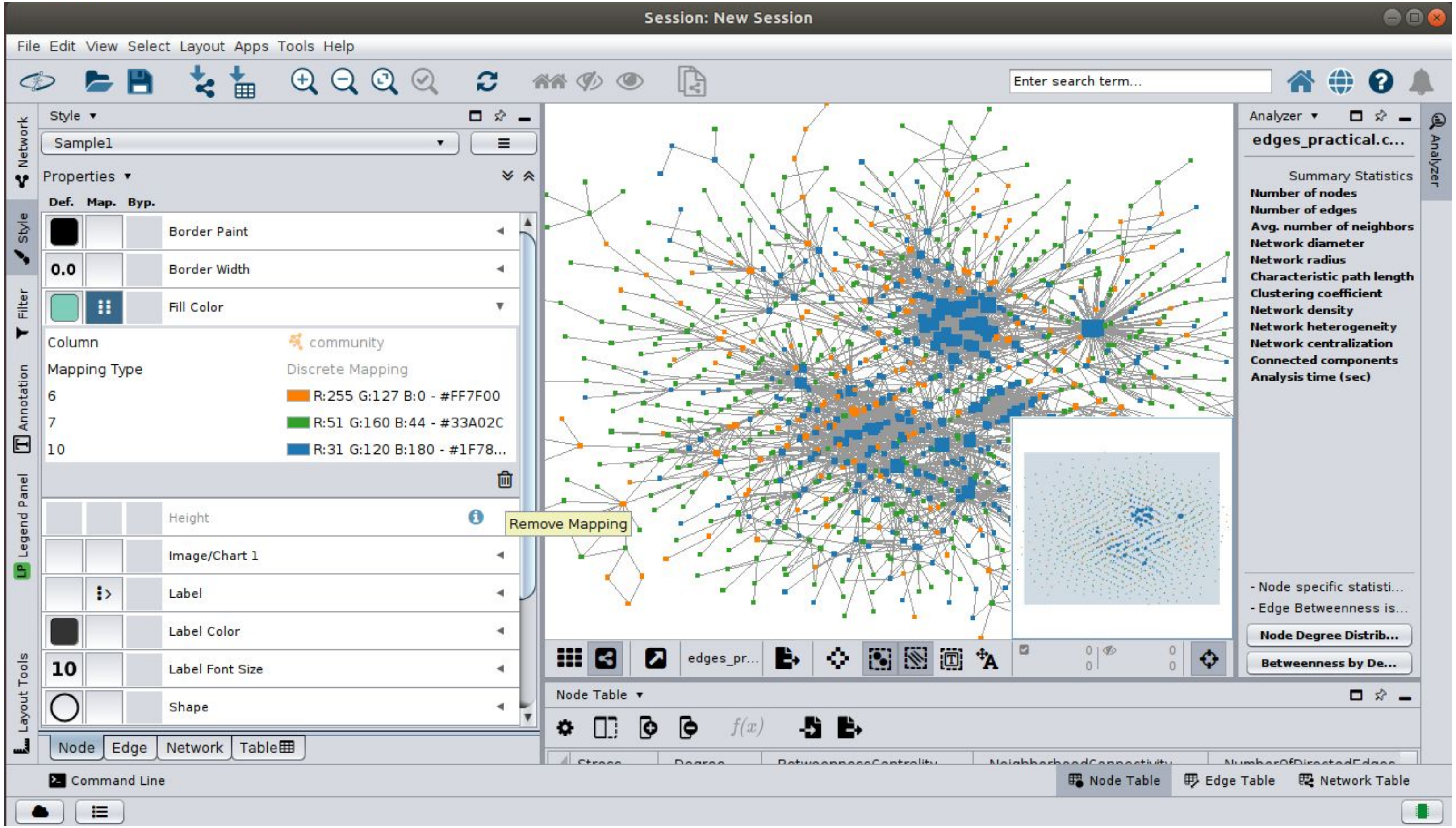
To save your image, go to option shown below and click on **Export as Image.**



Set your preferences as Landscape or Portrait. And also the file format of your image.



To visualize the betweeness centrality of nodes, first you need to remove the existing mapping of the node. Press **Remove Mapping** in the **Fill Color** option for **Node**.



To color the nodes according to its betweeness value, go to **Fill color** in **Styles**. Select **Column -> BetweenessCentrality**, **Mapping type -> Continuos.** To change color style or the range you can double click on the Current Mapping and modify it to suite your preferences.